CLINICAL PROGRAMMING BOOTCAMP • MODULE 05

Building SDTM Domains I

Special Purpose domains and DM

Your first real domain. One row per subject — and three different CRF forms feeding it.

Hands-on: the Excel mapping exercise, then Notebook 04 (SAS)

MODULE GOALS

By the end of this module you can...

1. **Explain what Special Purpose means** and why DM does not follow the three general observation classes.
2. **Build USUBJID correctly** and say why SUBJID alone is never enough.
3. **Derive AGE from a birth date** including the birthday edge case that catches everyone.
4. **Apply CT to SEX, RACE and ETHNIC** and know why blind upper-casing is dangerous in production.
5. **Populate the reference dates** and name which CRF form each one comes from.

RECAP AND PLACE

Where DM sits in the domain families

Interventions

Something given to the subject

`CM · EX`

Events

Something that happened

`AE · MH · DS`

Findings

Something measured

`VS · LB · EG`

Special Purpose

Its own fixed structure

`DM · CO · SE · SV`

Special Purpose means: this domain has its own defined structure

The three general observation classes share a common set of variables (a topic, timing variables, qualifiers, a --SEQ). Special Purpose domains do not follow that pattern — the SDTMIG defines their variables explicitly. DM is the one you will meet first and use most.

THE SHAPE

Two rules that define DM

1

One row per subject

8 subjects → exactly 8 rows. No subject appears twice, ever. This is the easiest domain to check: your row count must equal your subject count.

2

No --SEQ variable

There is no DMSEQ. Sequence numbers exist to tell a subject's multiple records apart — DM has only one record per subject, so it needs none.

DM is the spine of the study

Every other domain joins back to DM on USUBJID. DM also carries the reference dates that every --DY calculation in every other domain depends on. Build DM first, and build it right — everything downstream inherits its mistakes.

THE SURPRISE

DM is built from THREE forms, not one

1	2	3
Demographics	**Exposure**	**Disposition**
`dm_raw.csv`	`ex_raw.csv`	`ds_raw.csv`
Who the subject is: birth date, sex, race, ethnicity, country, arm, consent date, randomization date	RFSTDTC / RFENDTC — first and last dose. These define Study Day 1 for the whole study	RFPENDTC — the date the subject's participation ended

A common beginner assumption is that DM comes from the demographics form alone. It does not — and the reference dates it borrows are the most consequential values in the study.

THE MOST IMPORTANT VARIABLE

Building USUBJID

SUBJID is only unique within a site. USUBJID must be unique across the whole study — and identical for that subject in every domain.

The problem

```
Site 01, Subject 001
Site 02, Subject 001
```

Two different people.

The fix

```
ABC-01-01-001
ABC-01-02-001
```

Unambiguous.

The rule

```
USUBJID = STUDYID  ||  "-"  ||  SITEID  ||  "-"  ||  SUBJID
        = "ABC-01"    +      "01"     +      "001"      ->   ABC-01-01-001
```

Watch the leading zeros. If SITEID or SUBJID is read as a number, `01` becomes `1` and you build `ABC-01-1-1` — wrong for every subject, in every domain.

THE DERIVATION THAT CATCHES EVERYONE

AGE is calculated, not collected

The CRF collects a birth date. SDTM wants completed years at a defined reference date — here, informed consent.

Born	1969-05-14
Consented	2024-02-20
Simple year subtraction	2024 − 1969 = 55
Correct answer	54

Why 54?

Her birthday is 14 May. On 20 February 2024 it had not yet happened that year, so she had completed only 54 years.

The rule: subtract one if the birthday has not occurred by the reference date.

```
SAS:  intck("year", birth, ref, "C")
R:    completed-years helper
```

Which reference date? The protocol decides — consent or first dose. Using first dose here would make one of our subjects a year older. Never leave it implicit.

STANDARDIZING VALUES

CT in DM: SEX, RACE, ETHNIC, ARM

SEX	1 / 2	→ M / F	EDC stores a numeric code — you need the data dictionary to decode it
RACE	White · asian · 'White '	→ WHITE · ASIAN	Free text, mixed case, stray spaces — trim then normalise
ETHNIC	Not Hispanic or Latino	→ NOT HISPANIC OR LATINO	Upper-case to match CT exactly
ARMCD	Drug A / Placebo	→ A / P	Derived short code: max 20 chars, no spaces

Production warning. Blindly upper-casing free text happens to produce valid CT for this study. It would also let a site's "Caucasian" through un-flagged. Real mapping code uses an explicit lookup and fails loudly on anything it does not recognise.

PLANNED VERSUS ACTUAL

ARM, ARMCD, ACTARM, ACTARMCD

ARM / ARMCD

PLANNED — the arm the subject was randomized to.

Comes from the randomization on the demographics form.

ACTARM / ACTARMCD

ACTUAL — the arm the subject really received.

Determined from what was actually administered (EX).

In ABC-01 they are identical — every subject received the treatment they were randomized to, so ACTARM = ARM for all 8. That is the normal case, and it is why the copy looks trivial.

When would they differ? A subject randomized to Drug A who is dispensed placebo by a pharmacy error has ARM = Drug A but ACTARM = Placebo. Safety analyses use the actual arm; efficacy analyses usually use the planned one. All four variables are Required in SDTMIG v3.3.

TIMING ANCHORS

The six reference dates — and where each comes from

Variable	Meaning	Source	Note
RFSTDTC	First dose	ex_raw.EXSTDTC	*Defines Study Day 1 for the WHOLE study*
RFENDTC	Last dose	ex_raw.EXENDTC	
RFXSTDTC	First study treatment	ex_raw.EXSTDTC	
RFXENDTC	Last study treatment	ex_raw.EXENDTC	
RFICDTC	Informed consent	dm_raw.RFICDTC	*Also becomes a DS protocol milestone*
RFPENDTC	End of participation	ds_raw.EOSDT	*From the Disposition form*

Only two of the six come from the demographics form. Four are borrowed from other CRFs — which is why DM cannot be built in isolation.

THE TARGET

One finished DM row

USUBJID	SUBJID	SITEID	RFSTDTC	RFICDTC	AGE	AGEU	SEX	RACE	ARMCD	ARM	COUNTRY
ABC-01-01-001	001	01	2024-03-01	2024-02-20	54	YEARS	F	WHITE	A	Drug A	USA

(23 variables in total — 12 shown)

Copied as collected	SUBJID · SITEID · ARM · COUNTRY
Derived	USUBJID · AGE · ARMCD · ACTARMCD
Assigned	STUDYID · DOMAIN · AGEU
Mapped to CT	SEX · RACE · ETHNIC
Borrowed from another form	RFSTDTC · RFENDTC · RFXSTDTC · RFXENDTC · RFPENDTC

Every DM variable is one of these five kinds. Classify each one before you write any code — that IS the mapping specification.

LEARN FROM THESE

Common DM mistakes

Adding a DMSEQ
DM has no sequence variable — one row per subject means nothing to sequence.

USUBJID from SUBJID alone
Not unique across sites. Two subject 001s become one person.

Simple year subtraction for AGE
Gives 55 instead of 54. Check whether the birthday has occurred.

Blank RFSTDTC
You forgot to join EX. Every downstream --DY will be wrong or missing.

Omitting ACTARM / ACTARMCD
Both are Required in SDTMIG v3.3, even when they equal ARM.

Looking for end-of-study in DM raw
It is on the Disposition form — DM borrows RFPENDTC from it.

WHAT'S NEXT

Map it by hand, then automate it

1. **Step 1 · Excel** Map all 8 subjects into DM by hand in the mapping workbook. No code — just the rules.
2. **Step 2 · Notebook 04** Build the same DM dataset in SAS. Compare it to what you did by hand.
3. **Step 3 · Check** Compare against data/sdtm/dm.csv — the finished reference.

Doing it by hand first is deliberate: if you cannot map a row in a spreadsheet, you cannot write code that maps 800.